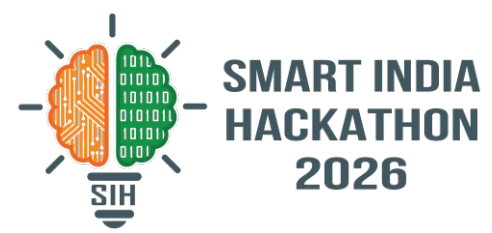


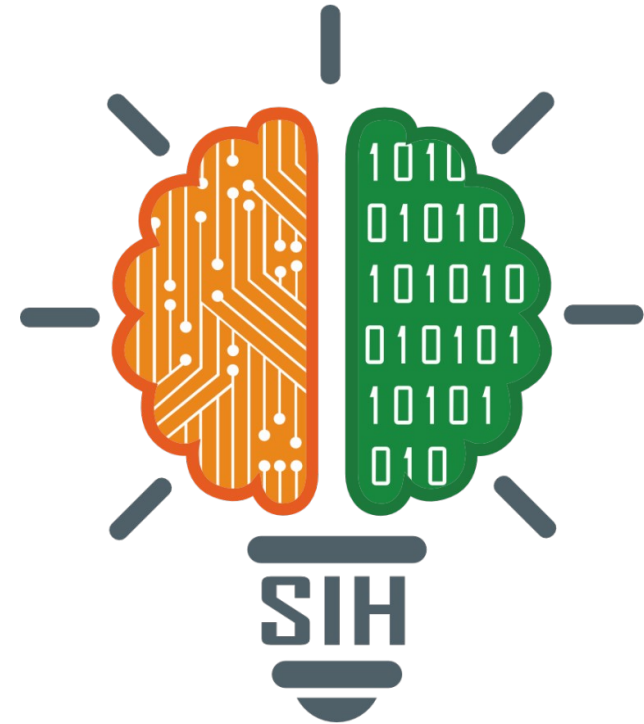


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26153
- **Problem Statement Title** – AI based Network Attack Forecasting from Network Traffic Data
- **Theme** – Blockchain and Cybersecurity
- **PS Category** – Software
- **Team ID** – 149775
- **Team Name (Registered on portal)** – Resurrección





SentiNet : A World Model that Forecasts Network Attacks



SMART INDIA
HACKATHON
2026

Enterprise & Critical Information Infrastructure | Learns $P(S_{t+1} | S_t)$ | K-step attack forecasting | MITRE ATT&CK stages | SHAP + attention explanations | Runs fully offline

IDEA / SOLUTION

Intrusion detectors tell you a breach has happened. **SentiNet tells you where the attacker is going next, while the attacker is still scanning and there is still time to stop them.**



Learns how the network changes over time

A world model learns $P(S_{t+1} | S_t)$: how a graph of hosts, flow and packet features changes each minute. It does not just label single flows.



Plays the future forward

Simulates 64 possible futures up to K windows ahead. The share that reach an infiltration state is the infiltration probability, shown with a confidence band.



Names the next attack stage and target

Each forecast maps to MITRE ATT&CK: Reconnaissance → Initial Access → Lateral Movement → C2 → Exfiltration, and names the host likely hit next.



Shows its reasoning

Attention weights and SHAP values list the flags, ports and flow statistics behind every forecast. No unexplained scores.



“What if we block it?”

The defender applies an action (block port 445, isolate a host) inside the model and sees how the risk changes, before touching the real network.



A working prototype, not only a plan

PCAP or CSV in; forecasts, Shapley explanations, flagged flows and signed receipts out. Offline app, ordinary CPU.

PROBLEM RESOLUTION

- **Classifiers see flows, not campaigns:** a benign/malicious label per flow throws away order and timing. A slow scan, then a brute-force, then lateral SMB traffic is one story, not three unrelated alerts.
- **Alerts come after the damage:** detection fires once a stage has already happened. Defenders need the warning while the attacker is still in reconnaissance.
- **Scores nobody can act on:** a probability with no “why” and no “what next” gets ignored by SOC analysts and cannot be audited in CII.

INNOVATION & UNIQUENESS



Attack “weather forecast”

Many sampled futures give a risk band and time-to-compromise, not one label.



Knows the attacker’s playbook

An ATT&CK/CAPEC knowledge base names the technique and its mitigation; CVE exposure raises host risk.



Flow + packet, on a graph

TTL, window size and port order catch slow scans; the graph predicts which host is next.



What-if defence simulator

Test a block or isolate action inside the model before using it.



Auditable forecasts

Each forecast is signed into a Merkle ledger and later scored against what actually happened.



Cheap enough for CII

~170k parameters: 12 h of traffic forecast in seconds on a laptop CPU, fully offline.

Built: flow + packet pipeline

PCAP (Scapy) + CIC / CTU-13 / UNSW loaders

Built: world model + forecaster

K-step rollouts, Shapley, what-if

Built: offline demo app

Streamlit + CLI + signed receipts

Next: real-data training

CIC-IDS2018, CTU-13, DARPA 2000; CII pilot



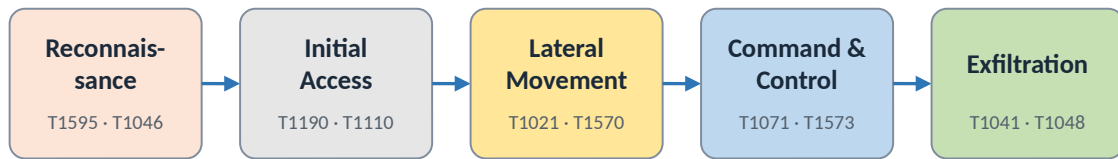
TECHNICAL APPROACH

WORLD MODEL ARCHITECTURE

SentiNet learns **how network state changes**. An encoder turns each time window into a state, a dynamics model learns $P(S_{t+1} | S_t)$, and a rollout engine simulates K steps ahead.

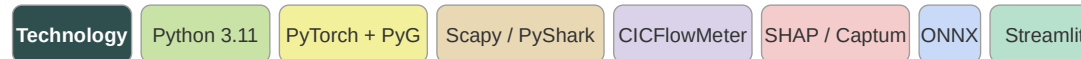
- **State S_t (every 60 s window):** a graph with hosts as nodes and flows as edges. Flow-level features: TCP flag counts (SYN/ACK/FIN/RST/PSH/URG), bytes, packets, duration, IAT mean/var/max, bidirectional ratio. Packet-level features: TTL mean and variance, TCP window size, fragment flags, payload-size histogram, sequential vs random port order, retransmissions.
- **Encoder:** edge-weighted GraphSAGE over the host graph + an MLP on the state vector $\rightarrow$ one latent vector per window, plus one per host.
- **Dynamics (the world model):** recurrent state-space model (GRU memory + stochastic latent) with a causal Temporal Transformer over the last 30 windows. It learns to predict the next state's features (self-supervised) and the next ATT&CK stage from the datasets' attack timelines (supervised dynamics learning).
- **Forward simulation:** sample 64 latent paths $K = 1 \dots 10$ windows ahead $\rightarrow$ $P(\text{infiltration within } K)$, predicted stage and next target host.
- **Explanations:** attention over time and neighbours + SHAP over feature groups $\rightarrow$ e.g. "top drivers: SYN-only ratio $\uparrow$, dst-port entropy $\uparrow$, TTL variance $\uparrow$ ".

FORECAST ATTACK STAGES (MITRE ATT&CK)

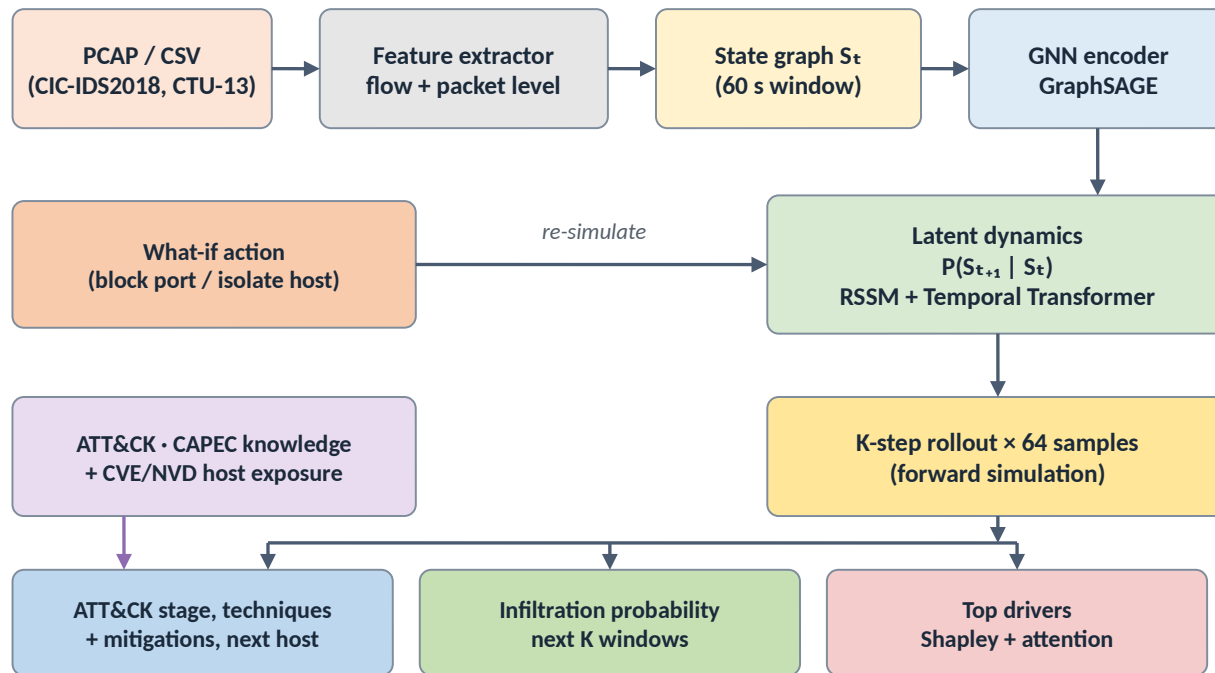


Dataset labels (e.g. CIC-IDS2018 "Infiltration", "Brute Force", "Bot"; DARPA 2000 LLDOS phases 1-5) are mapped to these stages. The benign $\rightarrow$ recon $\rightarrow$ access transitions are the ones the model must learn to see coming.

SENTINET TECH STACK



WORLD-MODEL PIPELINE



Every forecast is signed into our Merkle ledger, so it can be checked later against what actually happened.

Supervised dynamics learning: ground-truth transitions come from each dataset's attack timeline. The loss weights transitions into attack stages, so rare stages are learned. **Output per window:** $P(\text{infiltration in the next } K \text{ windows})$, predicted stage, top contributing flags/ports/flow stats, and the next target host. **Not a static classifier:** we show it by shuffling time. If accuracy does not drop, it has not learned dynamics.



FEASIBILITY & VIABILITY

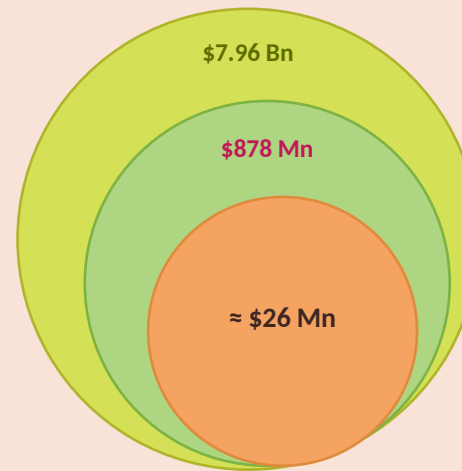
POTENTIAL CHALLENGES AND RISKS:

Attack windows are rare: attack time is a small fraction of each capture, and stages like exfiltration are rarer still.

Order may carry no signal: our earlier project found shuffling time changed nothing. Here we test it on every result: shuffling history drops F1 from 0.865 to 0.347.

Dataset shift: a model trained on a lab network may not carry over to a real enterprise or CII network.

Market Opportunity Funnel



TAM \$7.96 Bn

Global Intrusion Detection & Prevention market, 2026
(Precedence Research)

SAM up to \$878 Mn

India OT security market, 2026
(MarketsandMarkets)

SOM ≈ \$26 Mn

We aim for 3% of the available market: SOC's that want early warning, not only alerts.

STRATEGIES TO OVERCOME THESE CHALLENGES:

- **Weight the rare stages:** focal loss, oversampling of attack transitions, training on several datasets (CIC-IDS2018, CTU-13, UNSW-NB15, CICIOT2023).
- **Prove the time effect:** every result is shown next to logistic regression on the same features and next to our model with time shuffled. If there is no gap, we say so.
- **No-leak, unseen-attack tests:** split by time, never by random row. Hold out whole attack families and whole datasets. Fine-tune per site on ~10 days of its own benign traffic.

Component	Trained / tested on	What we will report
Feature pipeline · Scapy + CICFlowMeter	CIC-IDS2018 CSV + PCAP, CTU-13	Timestamped, normalised flow + packet matrix; flows/s
World model · RSSM + Transformer + GNN	CIC-IDS2018, CTU-13, UNSW-NB15	Next-state error; stage F1 at $t+1 \dots t+K$
Forecast engine · K-step rollout	DARPA 2000 LLDOS, LANL red team	Lead time before compromise; AUC@K; Brier score
Baseline · Logistic regression	Same features, same time split	F1, precision, recall, FPR, side by side
Generalisation	Leave-one-attack-out; CICIOT2023	Recall on attack types never seen in training
Explainer · SHAP + attention	All of the above	Deletion test: remove top-5 drivers → risk must fall

PROOF IT RUNS TODAY:

Runs on a laptop CPU: 12 h of traffic (67,650 flows) forecast in ~2 s: 720 windows x 64 simulated futures.

PCAP in, offline: a 441k-packet PCAP parsed into flow + packet features in 2.7 s. No GPU, no cloud.

Beats the baseline: F1 0.865 vs 0.483 (logistic regression) at the same 3.1% false-positive rate.*

Tested: 28 automated tests: loaders, PCAP round-trip, Shapley efficiency, tamper detection.

**Current scores are on held-out simulated scenarios (bundled simulator). This table is the real-data plan.*



IMPACT & BENEFITS

SENTINET VS. EXISTING TOOLS

Capability	Forecasts next stage	Learns time dynamics	ATT&CK stage mapping	Explains each output	Works offline
SentiNet	Yes: K-step probability	Yes: world model	Yes, per forecast	Yes: SHAP + attention	Yes
Snort / Suricata	No	No: rules	Partial (rule tags)	Rule name only	Yes
ML flow classifiers (RF / XGBoost IDS)	No	No: one flow at a time	No	Sometimes	Yes
Commercial NDR / SIEM (e.g. Darktrace, Splunk)	Risk scores, not stage forecasts	Varies	Yes	Partial	Varies: often cloud

IMPACTS

The blind spot: SOCs protecting India's power, banking, telecom and government networks see an attack only after a stage completes, and CERT-In's 6-hour reporting clock starts after the damage is done.

What changes: a forecast minutes to hours ahead, naming the next stage and the host likely hit next.

For operators: an early warning while the attacker is still scanning.

For analysts: one forecast carrying the whole chain and its drivers, not scattered alerts.

For auditors: every forecast is signed and later checked against reality, so accuracy itself is auditable.

For the nation: runs inside the premises of NCIIPC-notified CII. No traffic data leaves the site.

Beyond this: hospitals, rail signalling, smart grids and IoT (tested on CIIoT2023).

USE CASES



Power & Energy Grid (CII)

Forecasts recon → lateral movement towards SCADA hosts before control is reached.



Government SOC & CERT-In

ATT&CK-mapped forecasts and signed evidence fit the 6-hour reporting rule.



Banking & Finance

Predicts brute-force → access → exfiltration and names the server at risk.



Enterprises, MSMEs & Colleges

CPU-only, offline, one SPAN port: early warning they can afford.



Security Benefits

Act before compromise: block, isolate or watch while the attacker is still in recon.

Knows the next target: response is focused on the host most likely hit next.

Catches the unseen: judges stage behaviour, not signatures.



Economic Benefits

Cheaper to stop early: containing at recon costs far less than cleaning up a breach.

Low hardware cost: one CPU box per segment, no GPU, no licence fees.

Less alert fatigue: one forecast per campaign, not dozens of alerts.



Technical Benefits

Offline & open source: code, weights and training configs included, reproducible.

Two levels of traffic: flow + packet features fused on a graph.

What-if simulation: try a defensive action in the model first.



RESEARCH & REFERENCES

CITATIONS

World Models

Ha & Schmidhuber, 2018

Learn an internal model, then roll it forward.

Latent Dynamics (RSSM)

Hafner et al., DreamerV3, 2023

The recurrent state-space model we adapt to network state.

Graph NIDS

Lo et al., E-GraphSAGE, IEEE NOMS 2022

Flows as graph edges; basis of our encoder.

Attack Forecasting

Husák et al., IEEE COMST 2019

Survey of attack projection, prediction & forecasting.

Explainability

Lundberg & Lee, NeurIPS 2017 (SHAP)

Feature attribution for every forecast.

LINKS (Ctrl + Click on the blue words)



[Demo Video - Prototype](#)



[GitHub Repository](#)



[Documentation](#)



[Presentation](#)

SentiNet • Standards & Knowledge Alignment

MITRE ATT&CK

Stage labels + technique IDs for every forecast (T1595, T1190, T1021, T1071, T1041...).

CAPEC

Attack patterns linked to every ATT&CK technique the evidence points to.

CVE / NVD

CVSS exposure of each host's open services raises its chance of being next.

NCIIPC • CII

Design fits NCIIPC guidance for protected systems: on-prem, passive, no data leaves.

CERT-In

ATT&CK-mapped, signed forecasts support the 6-hour incident reporting rule.

RFC 6962

Merkle transparency log (as used for web certificates) for tamper-evident forecast receipts.

DATASET IMPLEMENTATION

1. CIC-IDS2017/2018: flows + PCAP, day-wise attack timelines

2. CTU-13: 13 botnet scenarios, labelled NetFlow + PCAP

3. UNSW-NB15: 9 attack families, PCAP + flows

4. CICIOT2023: 33 IoT attacks (unseen-attack tests)

5. DARPA 2000 LLDOS 1.0/2.0: 5-phase multi-stage attack

6. LANL auth + red team: real lateral movement, 58 days

7. MITRE ATT&CK • CAPEC • CVE/NVD: knowledge base

8. Own capture, IPv4 + IPv6: benign baseline (already collected)

MODEL SELECTION — HEAD-TO-HEAD

Component	Compared against	Pass condition
World model • RSSM + Transformer	Logistic regression; same model with history shuffled	Done* : F1 0.865 vs 0.483 at equal FPR; shuffled history drops to 0.347
Graph encoder • GraphSAGE	Flat feature vector	Better next-target accuracy from who-talks-to-whom
Rollout • 64 samples	Single one-step prediction	Calibrated probabilities (Brier / ECE)
Explainer	Attention only • SHAP • Integrated Gradients	Faithful : removing top drivers drops the risk
ATT&CK / CAPEC knowledge	Stage label only	Actionable : each alarm names technique, CAPEC pattern and mitigation

*Measured on held-out simulated scenarios (results/benchmark.md).
Other rows: planned, same features and time split, published win or lose.